

Image-Space Residual INR: Follow-up Pixel-Supervision Sweep

Patient-Specific Residual INR — slam (change_extent=0)

July 8, 2026

Setup

The current (follow-up) image is reconstructed in image space as $\hat{x} = f_{\theta}^{\text{prior}} + r_{\phi}$, with the prior INR $f_{\theta}^{\text{prior}}$ fit to the registered DICOM prior and frozen. The residual r_{ϕ} is trained using only a fraction p of the follow-up pixels (chosen uniformly at random) and evaluated on the full image. Baselines: **prior-copy** (DICOM prior, uses 0% of the follow-up) and **prior-finetune** (NeRP-style: the prior INR itself, initialized on the DICOM, fine-tuned on the same observed follow-up pixels).

Table 1: Reconstruction of the full follow-up image vs. supervision fraction p (higher PSNR/SSIM better; lower NMSE/CPE better; PBS ≈ 1 preserves change, $\ll 1$ suppresses it).

p (pixels)	method	PSNR	SSIM	NMSE	CPE	PBS
25%	prior-copy	24.22	0.826	0.0707	0.0268	0.000
	prior-finetune (NeRP)	29.03	0.888	0.0234	0.0156	1.093
	prior+residual (ours)	29.33	0.860	0.0218	0.0162	0.887
50%	prior-copy	24.22	0.826	0.0707	0.0268	0.000
	prior-finetune (NeRP)	32.08	0.930	0.0116	0.0100	1.046
	prior+residual (ours)	31.34	0.897	0.0137	0.0124	0.941
75%	prior-copy	24.22	0.826	0.0707	0.0268	0.000
	prior-finetune (NeRP)	35.55	0.961	0.0052	0.0059	1.020
	prior+residual (ours)	34.20	0.908	0.0071	0.0094	0.979
100%	prior-copy	24.22	0.826	0.0707	0.0268	0.000
	prior-finetune (NeRP)	49.02	0.989	0.0002	0.0022	1.003
	prior+residual (ours)	36.68	0.919	0.0040	0.0077	0.988

Table 2: How well the learned residual recovers the true interval change ($|x_{\text{ref}}| - |x_{\text{prior}}|$) for the ours model.

p (pixels)	corr	rel-L1
25%	0.818	0.638
50%	0.890	0.493
75%	0.943	0.385
100%	0.967	0.321

Prior INR reconstruction

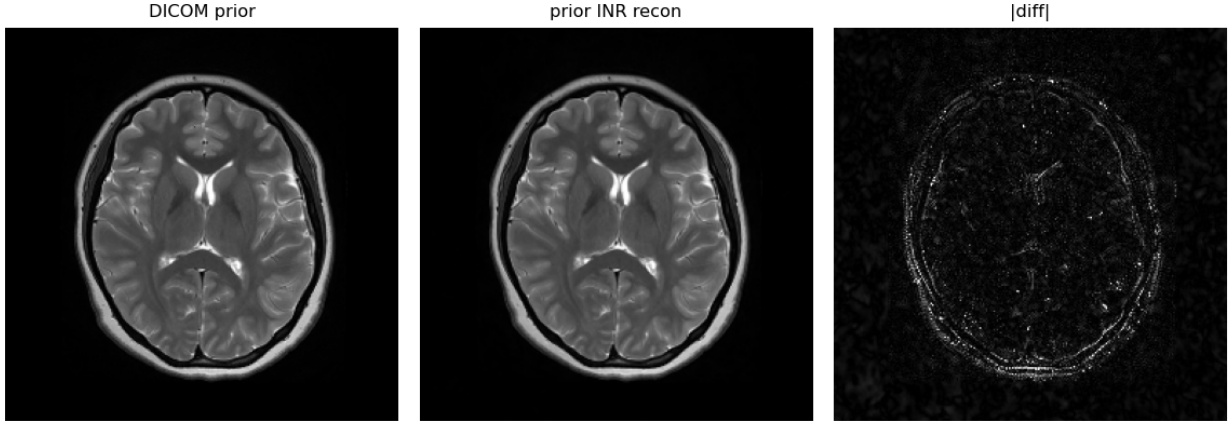


Figure 1: The frozen prior INR and the DICOM image it reconstructs.

Qualitative results (ours) per supervision fraction

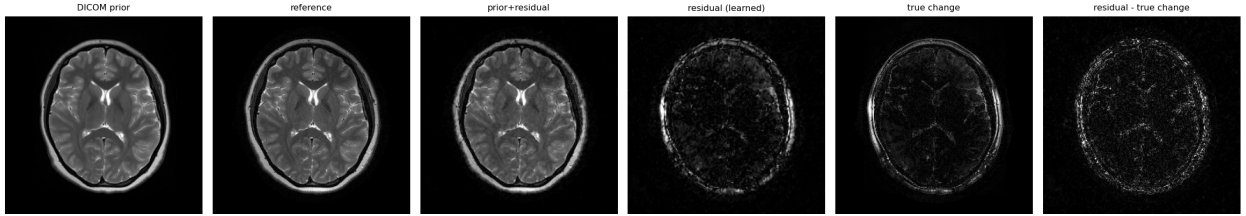


Figure 2: $p = 25\%$ of follow-up pixels observed. Left to right: DICOM prior, reference follow-up, our reconstruction ($f^{\text{prior}} + r$), learned residual r , true change, and the error $|r - \text{true change}|$.

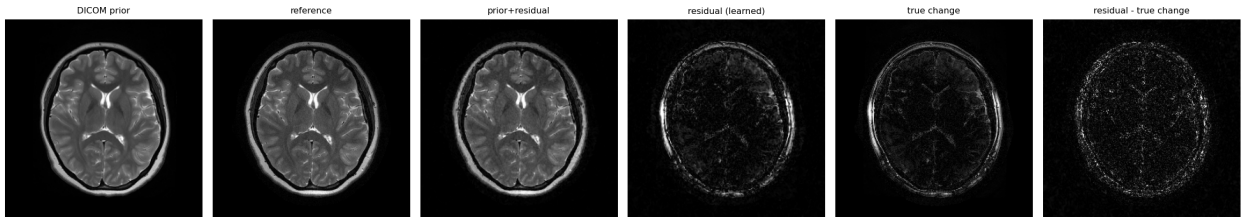


Figure 3: $p = 50\%$ of follow-up pixels observed. Left to right: DICOM prior, reference follow-up, our reconstruction ($f^{\text{prior}} + r$), learned residual r , true change, and the error $|r - \text{true change}|$.

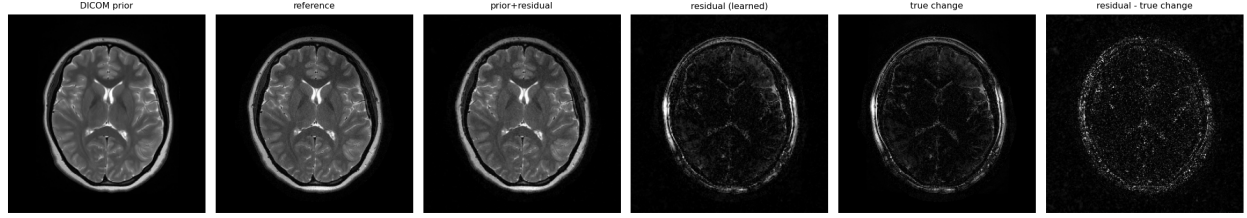


Figure 4: $p = 75\%$ of follow-up pixels observed. Left to right: DICOM prior, reference follow-up, our reconstruction ($f^{\text{prior}} + r$), learned residual r , true change, and the error $|r - \text{true change}|$.

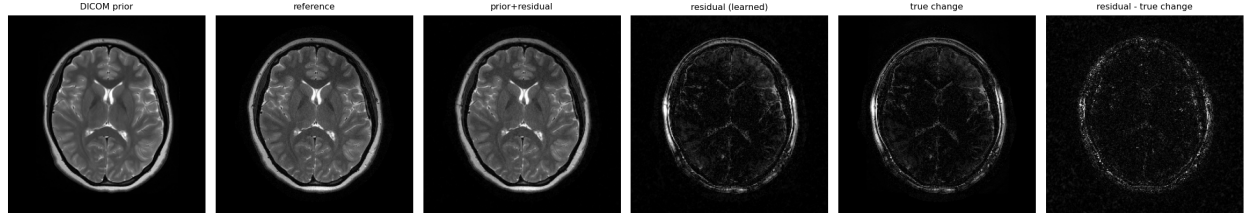


Figure 5: $p = 100\%$ of follow-up pixels observed. Left to right: DICOM prior, reference follow-up, our reconstruction ($f^{\text{prior}} + r$), learned residual r , true change, and the error $|r - \text{true change}|$.